Differentiation and Integration

1. Hardy-Littlewood maximal function

Def V.1.1. If f is integrable on $\mathbb{R}^d$. define its maximal function f^* by.

$$f^*(x) = \sup_{x \in B} \frac{1}{m(B)} \int_B |f(y)|\, dy \quad , \quad x \in \mathbb{R}^d.$$ where the supremum is taken over all balls containing x

Thm V.1.2 f is integrable on $\mathbb{R}^d$. then (i) f^* is measurable;

(ii) $f^*(x) < \infty$. for a.e. x.

(iii) $m(\{f^* > \alpha\}) \le \frac{3^d}{\alpha} \|f\|_{L^1(\mathbb{R}^d)}$ for all $\alpha > 0$, $\|f\| = \int_{\mathbb{R}^d} |f(x)|\, dx$.

(the weak type estimate for the maximal function).

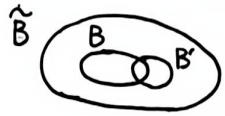
Proof: (i) $E_\alpha = \{f^* > \alpha\}$. If $\bar{x} \in E_\alpha$, $\exists$ ball B. s.t. $\bar{x} \in B$. $\frac{1}{m(B)} \int_B |f(y)|\, dy > \alpha$

Any point x close enough to $\bar{x} \in B \Rightarrow x \in E_\alpha$ thus E_α is open thus measurable

(ii) is the consequence of (iii).

Lemma: $\mathcal{B} = \{B_1, \dots, B_N\}$. open balls in $\mathbb{R}^d$. Then $\exists$ disjoint $\{B_{i_1}, B_{i_2}, \dots, B_{i_k}\} \subseteq \mathcal{B}$ s.t. $m\left(\bigcup_{l=1}^N B_l\right) \le 3^d \sum_{j=1}^k m(B_{i_j})$

Pf: If $B \cap B' \neq \phi$ $r(B) \ge r(B')$. Then $B' \subseteq \widetilde{B}$. where $\widetilde{B} = 3B$.



Choose B_{i_1} with maximal radius.

delect all $B_k \in \mathcal{B}$ with $B_k \cap B_{i_1} \neq \phi$

Then $B_k \subseteq \widetilde{B}_{i_1}$. where $\widetilde{B}_{i_1} = 3B_{i_1}$.

Continue doing this way, then we have $B_{i_1}, \dots, B_{i_k}$ and following $\widetilde{B}_{i_j}$.

$$\Rightarrow m\left(\bigcup_{l=1}^N B_l\right) \le m\left(\bigcup_{j=1}^k \widetilde{B}_{i_j}\right) \le \sum_{j=1}^k m(\widetilde{B}_{i_j}) = 3^d \sum_{j=1}^k m(B_{i_j})$$

Back to the proof, $E_\alpha = \{f^* > \alpha\}$.

$\forall x \in E_\alpha$. $\exists B_x$ containing x s.t. $\frac{1}{m(B_x)} \int_{B_x} |f(y)|\, dy > \alpha$.

Therefore, $\forall B_x$ $m(B_x) < \frac{1}{\alpha} \int_{B_x} |f(y)|\, dy$

$\forall K \subseteq E_\alpha$ K compact. $\exists$ finite-subcover of K, say $K \subseteq \bigcup_{l=1}^N B_l$.

$\exists$ disjoint $B_{i_1}, \dots, B_{i_k}$ s.t. $m\left(\bigcup_{l=1}^N B_l\right) \le 3^d \sum_{j=1}^k m(B_{i_j})$.

$$\Rightarrow m(K) \le \frac{3^d}{\alpha} \int_{\mathbb{R}^d} |f(y)|\, dy \quad \text{for all } K.$$

By approximating E_α with K. we have $m(E_\alpha) \le \frac{3^d}{\alpha} \int_{\mathbb{R}^d} |f(y)|\, dy$ $\square$.

2. Differentiation, Average problem

Thm V.2.1. f is integrable on $\mathbb{R}^d$ then $\lim\limits_{\substack{m(B)\to 0\\ x\in B}} \frac{1}{m(B)}\int_B f(y)dy = f(x)$ for a.e. x

Pf: Let $E_\alpha = \{x \mid \limsup\limits_{\substack{m(B)\to 0\\ x\in B}} \left|\frac{1}{m(B)}\int_B f(y)dy - f(x)\right| > 2\alpha\}$

We suffices to show that $m(E_\alpha)=0$

$\forall \varepsilon>0$ $\exists$ continuous function g of compact support with $\|f-g\|_{L^1}<\varepsilon$.

Then $\lim\limits_{\substack{m(B)\to 0\\ x\in B}} \frac{1}{m(B)}\int_B (f(y)-g(x))dy \le \lim\limits_{\substack{m\to 0\\ x\in B}} \frac{1}{m(B)}\left(\int_{B\cap \mathrm{supp}(g)} (g(y)-g(x))dy + \int_{B\cap \mathrm{Supp}(g)^c}\cdots\right)$

$B\cap \mathrm{supp}(g)$ is compact $\Rightarrow g|_{B\cap \mathrm{supp}(g)}$ is uniformly continuous

$\Rightarrow \exists \delta>0$ when $|y-x|<\delta$ $|g(y)-g(x)|<\varepsilon$. $\Rightarrow \lim \frac{1}{m(B)}\int_B g(y)dy = g(x)$

Then $\limsup\limits_{\substack{m(B)\to 0\\ x\in B}} \left|\frac{1}{m(B)}\int_B f(y)dy - f(x)\right| \le (f-g)^*(x) + |g(x)-f(x)|$

$F_\alpha = \{(f-g)^* > \alpha\}$. $G_\alpha = \{|f(x)-g(x)|>\alpha\}$. $\Rightarrow E_\alpha \subseteq F_\alpha \cup G_\alpha$

By Chebychev's inequality: $m(G_\alpha) \le \frac{1}{\alpha}\|f-g\|_{L^1}$ $\left(\chi_{G_\alpha} \le \frac{1}{\alpha}|f-g|\right)$

By weak type estimate for the max function $m(F_\alpha) \le \frac{3^d}{\alpha}\|f-g\|_{L^1}$

$\Rightarrow m(E_\alpha) \le \frac{A}{\alpha}\varepsilon + \frac{1}{\alpha}\varepsilon$ $\square$.

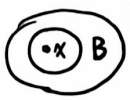
Def V.2.2. f measurable, f is locally integrable iff $\forall$ ball B, we have $f\chi_B$ is integrable

Denote $L^1_{loc}(\mathbb{R}^d)$ the space of all locally integrable functions.

Thm V.2.2. If $f\in L^1_{loc}(\mathbb{R}^d)$. then $\lim\limits_{\substack{m(B)\to 0\\ x\in B}} \frac{1}{m(B)}\int_B f(y)dy = f(x)$ for a.e. x.

Def V.2.3. (1) E measurable, $x\in\mathbb{R}^d$. We say x is a point of Lebesgue density of E if

$\lim\limits_{\substack{m(B)\to 0\\ x\in B}} \frac{m(B\cap E)}{m(B)} = 1$



(2) $f\in L^1_{loc}(\mathbb{R}^d)$. the Lebesgue set of $f := \{\bar{x} \mid f(\bar{x})<\infty,\ \lim\limits_{\substack{m(B)\to 0\\ \bar{x}\in B}} \frac{1}{m(B)}\int_B |f(y)-f(\bar{x})|dy = 0$

Cor V.2.4. Suppose E measurable, Then: (i) Almost every $x\in E$ is a point of density of E

(ii) Almost every $x\notin E$ is not a point of density of E.

Pf: Take $f=\chi_E$ then $\frac{m(B\cap E)}{m(B)} = \frac{1}{m(B)}\int_B \chi_E(x)dx \to 1$ $x\in E$

$\to 0$ $x\notin E$. $\square$.

Coro V.2.5. $f \in L^1_{loc}(\mathbb{R}^d)$ then almost every point belongs to the Lebesgue set of f.

Proof: $\forall r \in \mathbb{Q} \quad \exists E_r, m(E_r) = 0$. s.t. $\lim_{\substack{m(B)\to 0 \\ x\in B}} \frac{1}{m(B)} \int_B |f(y) - r| dy = |f(x) - r| \quad \forall x \notin E_r$.

Let $E = \bigcup_{r\in\mathbb{Q}} E_r$, $m(E) = 0 \quad \forall \bar{x} \in E$. $\exists r \in \mathbb{Q}$ s.t. $|f(\bar{x}) - r| < \varepsilon$.

$\Rightarrow \frac{1}{m(B)} \int_B |f(y) - f(\bar{x})| dy \leq \frac{1}{m(B)} \int_B |f(y) - r| dy + |f(\bar{x}) - r|$

$\Rightarrow \limsup \frac{1}{m(B)} \int_B |f(y) - f(\bar{x})| dy \leq 2\varepsilon$. $\Rightarrow \bar{x} \in$ lebesgue set of f.

3. Functions of bounded variation